

# LINJIE LYU

✉ llyu@mpi-inf.mpg.de

🌐 <https://linjielyu.github.io/>

## RESEARCH INTERESTS

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- Diffusion Models Based 2D/3D Generation and Editing
- Global-illumination-aware Inverse Rendering, 3D Reconstruction, Relighting
- Radiance Fields, Efficient 3D Gaussian Splatting
- Uncertainty Quantification

## EDUCATION

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**Max-Planck-Institut für Informatik**, Saarbrücken, Germany  
Ph.D. Student in Computer Graphics and Vision.

June 2021 — January 2026

- **Inverse Rendering & 3D Reconstruction:** Built frameworks to extract geometry, materials, and complex lighting from 2D images by integrating global illumination into differentiable pipelines.
- **Rendering Efficiency:** Replaced expensive Monte Carlo integration with highly efficient alternatives, including spherical harmonics for soft shadows and Neural Precomputed Radiance Transfer (PRT) fields.
- **Generative Model Integration:** Leveraged diffusion models (DDPMs) for ambiguity resolution in inverse problems and developed intrinsic-image latent space workflows for realistic scene editing.
- **Probabilistic Modeling:** Introduced uncertainty-aware rendering and differentiable manifold sampling to optimize subsequent data capture (next-best viewpoint planning).

**Universität des Saarlandes**, Saarbrücken, Germany  
Graduate School in Computer Science

October 2019 — March 2021

**Tsinghua University**, Beijing, China  
Bachelor in Physics and Mathematics

September 2015 — July 2019

\* College Scholars; University Honors

**Sichuan Mianyang High School**, Mianyang, China  
\* Silver Medal, Chinese Physics Olympiad

September 2012 — June 2015

## JOBS

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**Bridge (LinkedIn)**, Frankfurt, Germany  
Co-Founder & CTO

January 2026 — Present

Co-founded and lead the technical development of an AI-powered buyer platform enabling Western companies to source directly from verified Chinese factories without the need for local teams. Architecting AI-driven solutions to streamline supplier selection, bridge complex technical communication gaps, and enforce rigorous quality control specifications.

## INTERSHIP

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**Adobe**, London, UK

June 2024 — September 2024

Research Intern During my research internship at Adobe, I developed a generative framework for pixel-precise image editing within an intrinsic-image latent space, culminating in our SIGGRAPH 2025 publication, IntrinsicEdit. This pipeline enables semantic, local manipulations—such as object insertion and removal, material adjustments, and scene relighting—while automatically resolving complex global-illumination effects like reflections and shadows.

CUDA-based visibility computation in Differentiable Rendering Tool ([Github](#))

Engineered the core CUDA/C++ forward visibility and occlusion computation module for a custom differentiable rasterizer used in top-tier computer vision publications. Designed high-performance kernels to compute visibility independently of external graphics APIs (e.g., OpenGL) and integrated the module as a custom TensorFlow operator to support multi-view batch rendering and parametric shape fitting.

## PUBLICATIONS

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- **L. Lyu**, A. Tewari, J. Chen, T. Leimkühler, C. Theobalt. *Faster 3D Gaussian Splatting Convergence via Structure-Aware Densification*. Conference Proceedings, SIGGRAPH 2026
- X. Zhou, J. Chen, P. Rao, T. Teufel, **L. Lyu**, T. Minasian, O. Sotnychenko, X.-X. Long, M. Habermann, C. Theobalt. *OLATverse: A Large-scale Real-world Object Dataset with Precise Lighting Control*. CVPR 2026 (Oral).
- **L. Lyu**, V. Deschaintre, Y. Hold-Geoffroy, M. Hasan, J. Yoon, T. Leimkühler, C. Theobalt, I. Georgiev. *IntrinsicEdit: Precise Generative Image Manipulation in Intrinsic Space*. ACM Transactions on Graphics, SIGGRAPH 2025.
- **L. Lyu**, A. Tewari, M. Habermann, S. Saito, M. Zollhöfer, T. Leimkühler, C. Theobalt. *Manifold Sampling for Differentiable Uncertainty in Radiance Fields*. Conference Proceedings, SIGGRAPH Asia 2024
- **L. Lyu**, A. Tewari, M. Habermann, S. Saito, M. Zollhöfer, T. Leimkühler, C. Theobalt. *Diffusion Posterior Illumination for Ambiguity-aware Inverse Rendering*. ACM Transactions on Graphics, SIGGRAPH Asia 2023.
- **L. Lyu**, A. Tewari, T. Leimkühler, M. Habermann, C. Theobalt. *Neural Radiance Transfer Fields for Relightable Novel-view Synthesis with Global Illumination*. ECCV 2022 (Oral).
- **L. Lyu**, M. Habermann, L. Liu, M. B. R, A. Tewari, C. Theobalt. *Efficient and Differentiable Shadow Computation for Inverse Problems*. ICCV 2021.

## SERVICES

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**Reviewer**

SIGGRAPH, SIGGRAPH Asia, Eurographics, Pacific Graphics, 3DV

**Teaching**

Computer Vision and Machine Learning for Computer Graphics